



Modules 2 assi:4

Cyber security

1. Connect a USB webcam to your computer, install any required drivers, and verify that the webcam works by opening the default camera app and capturing a photo

Practical: Connect and Test a USB Webcam

Connect the webcam – Plug the USB webcam into an available USB port on the computer.

Install drivers – Wait for Windows to automatically install the required driver. If prompted, install the driver provided by the webcam manufacturer.

Check the webcam – Open **Device Manager** → **Cameras** and confirm that the webcam is listed without a warning symbol.

Open the Camera app – Click **Start** → **Camera** to open the default Windows Camera application.

Select the webcam – If more than one camera is available, select the connected USB webcam as the camera source.

Test the video – Check that a live video preview appears and that the image is clear.

Capture a photo – Click the **Take Photo** button in the Camera app and confirm that the photo is saved successfully.

Verify the result – Open the saved photo to confirm that the webcam is working correctly.

2. Set up a local printer (USB or network), install the correct driver, and print a test page to confirm successful installation.

Practical: Setting Up and Testing a Local Printer

Connect the printer – Connect the printer to the computer using a USB cable or connect it to the same Wi-Fi/network as the computer.

Turn on the printer – Switch on the printer and make sure it has paper and ink/toner.

Open Printer Settings – Go to Start → Settings → Devices → Printers & scanners.

Add the printer – Click Add a printer or scanner and select the detected printer.

Install the correct driver – Allow Windows to install the recommended driver, or install the correct driver from the printer manufacturer's driver package if required.

Set as default printer – Select the printer and choose Manage → Set as default if desired.

Print a test page – Open Printer Properties and select Print Test Page.

Verify installation – Check that the test page prints correctly, confirming that the printer, driver, and connection are working successfully.

3. Simulate a connectivity issue by unplugging the printer's USB cable, then use Windows Device Manager to identify and resolve the 'printer not detected' problem.

Hint: Check for error icons and use the 'Scan for hardware changes' option after reconnecting.

Practical: Troubleshoot a Printer Not Detected Using Device Manager

Disconnect the printer – Turn off the printer and unplug its USB cable from the computer.

Check Device Manager – Right-click Start → Device Manager and look under Printers, Print queues, or Universal Serial Bus controllers.

Observe the device status – Check for a yellow warning icon or a missing printer entry, which indicates that Windows may not be detecting the device correctly.

Reconnect the printer – Plug the USB cable back into the computer and turn the printer on.

Scan for hardware changes – In Device Manager, click Action → Scan for hardware changes. Windows should detect the reconnected printer.

Check the driver – If the printer appears with a warning icon, right-click it and select Update driver or Uninstall device, then reconnect the printer so Windows can reinstall the driver.

Verify the printer – Open Settings → Devices → Printers & scanners and confirm that the printer is listed and ready.

Print a test page – Send a test page to confirm that the printer is detected and working correctly.

4. A user reports that their webcam video is not displaying in Zoom, but it works in the default camera app. Troubleshoot and resolve the issue so the webcam works in both applications.

Hint: Check if another app is using the webcam or if Zoom permissions are enabled.

Troubleshooting: Webcam Works in Camera App but Not in Zoom

Verify the webcam works

Open the Windows Camera app.

Confirm that the webcam displays video correctly.

Close other applications using the webcam

Exit applications such as Camera, Microsoft Teams, Skype, Google Meet, or OBS Studio.

These applications may be using the webcam and preventing Zoom from accessing it.

Check Zoom camera settings

Open Zoom.

Go to Settings → Video.

Select the correct webcam from the Camera drop-down list.

Verify that a video preview appears.

Check Windows camera permissions

Open Settings → Privacy → Camera (or Privacy & Security → Camera).

Turn on Camera access.

Turn on Allow apps to access your camera.

Turn on Allow desktop apps to access your camera so Zoom can use the webcam.

Restart Zoom and the webcam

Close Zoom completely.

Disconnect and reconnect the webcam if it is an external USB camera.

Reopen Zoom and test the camera again.

Troubleshooting: Webcam Works in Camera App but Not in Zoom

Update drivers and Zoom

Open Device Manager → Cameras.

Right-click the webcam and select Update driver.

Install any available Zoom updates.

Restart the computer

Reboot the system to release any camera resources that may be locked by another application.

Test the webcam in Zoom

Start or join a Zoom meeting.

Click Start Video.

Confirm that the webcam video is displayed properly.

The background features a dark blue field on the right and a light blue field on the left, separated by a diagonal line. A thin, dark blue line runs parallel to the diagonal line, and a thin, light blue line runs parallel to the dark blue line.

Thank You